Statistical Inference

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February 27, 2026

Abstract

Notes for the basics of Probability Theory.

Contents

1 Fundamentals of Statistics

1.1 Random Samples and Statistics

Definition 1.1.1 (Random Samples)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be random variables. We say that they are a random sample of size n from the population f if the following are true.

- $X_1, \dots, X_n$ are pairwise independent.
- The probability density (mass) function is given by $f_{X_i} = f_X$.

The fact that their probability density functions are equal imply that the random variables are identically distributed.

Definition 1.1.2 (Statistics)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. A statistic of the random sample is a random variable $T(X_1, \dots, X_n)$ for some function $T : \mathbb{R}^n \to \mathbb{R}$.

Definition 1.1.3 (Sample Mean)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Define the sample mean of the random sample to be the statistic defined by

$$\overline{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

By the weak law of large numbers, we know that $\overline{X}$ converges in mean square to the mean of the random sample.

Definition 1.1.4 (Sample Variance)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Define the sample variance of the random sample to be the statistic defined by

$$S^2 = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \overline{X})^2$$

Definition 1.1.5 (Sample Standard Deviation)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Define the sample standard deviation of the random sample to be the statistic defined by

$$S = \sqrt{S^2}$$

2 Principles of Data Reduction

2.1 Sufficiency Principle

Suppose that $(\Omega, \mathcal{F}, P)$ is a probability space. Often we do not know what P is actually like. To probe the properties of P, we may use statistics $T(X_1, \dots, X_n)$ on the probability space.

A sufficient statistic T is a statistic where, informally speaking, all information about P must come from T.

Definition 2.1.1 (Sufficient Statistics)
Let $(\Omega, \mathcal{F}, P)$ be a collection of probability spaces for $\theta \in I$. Let $\{D_\theta \mid \theta \in I\}$ be a class of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. Let T be a statistic of the random sample. We say that T is sufficient if the conditional probability density function $f_{X_\theta | T}$ is a constant function with respect to θ.

Example 2.1.2
Let $\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample of Bernoulli distribution with parameter $\theta \in (0, 1)$ (Note that for each θ, $P_{X_i} = f_{X_i}$ is different since the formula for f_{X_i} depends on θ). Then $T = \sum_{i=1}^n X_i$ is a sufficient statistic for θ.

Proof
Let $\theta \in \mathbb{R}$. We use the above lemma. We have

$$\begin{aligned} P_\theta(\mathbf{X} = \mathbf{x} \mid T(\mathbf{X}) = T(\mathbf{x})) &= \frac{f_{X_1,\dots,X_n}(x_1, \dots, x_n)}{f_{T(X_1,\dots,X_n)}(x_1, \dots, x_n)} \\ &= \frac{\prod_{i=1}^n f_{X_i}(x_i)}{f_{X_1+\dots+X_n}(x_1, \dots, x_n)} \\ &= \frac{\prod_{i=1}^n \theta^{x_i}(1-\theta)^{1-x_i}}{\binom{n}{\sum_{i=1}^n x_i}\theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}} \\ &= \frac{\theta^{\sum_{i=1}^n x_i}(1-\theta)^{\sum_{i=1}^n x_i}}{\binom{n}{\sum_{i=1}^n x_i}\theta^{\sum_{i=1}^n x_i}(1-\theta)^{n-\sum_{i=1}^n x_i}} \\ &= \frac{1}{\binom{n}{\sum_{i=1}^n x_i}} \end{aligned}$$

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Theorem 2.1.3 (Neyman-Fisher Factorization Theorem)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. Let T be a statistic of the random sample. Let f_θ be the probability density function of X_θ. Then T is sufficient if and only if there exists probability density functions g_θ for each θ such that

$$f_\theta(x) = g_\theta(T(x))h(x)$$

for all $x \in \Omega$ and all $\theta \in I$.

Definition 2.1.4 (Minimal Sufficient Statistics)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. Let T be a sufficient statistic of the random sample. We say that T is minimal if for any other sufficient statistic T', $T'(x) = T'(y)$ implies $T(x) = T(y)$.

Definition 2.1.5 (Ancillary Statistics)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$

be a random sample with distribution D_θ. Let T be a statistic of the random sample. We say that T is ancillary if T is a constant function with respect to θ.

Definition 2.1.6 (Complete Statistics)

Theorem 2.1.7 (Basu's Theorem)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. Let T be a statistic of the random sample. If T is complete and minimal sufficient, then T is independent of any ancillary statistic.

2.2 Likelihood Principle

Definition 2.2.1 (The Likelihood Function)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. Define the likelihood function $L : I \to \{g : \mathbb{R} \to \mathbb{R} \mid g \text{ is a function }\}$ of the random sample to be

$$L(\theta) = f_\theta$$

where f_θ is the probability density (mass) function of X_θ.

2.3 Conditionality Principle

2.4 Equivariance Principle

3 Estimators

Suppose some real life problem is modelling statistically by the probability space $(\Omega, \mathcal{F}, P)$. To probe what P looks like, we sample the space using random variables $X_1, \dots, X_n : \Omega \to \mathbb{R}$. Based on the sample, we may infer that P may have parameter $\widehat{\theta}$ (naturally $\widehat{\theta}$ should be a function of $X_1, \dots, X_n$). We may then need to evaluate how accurate $\widehat{\theta}$ is compared to the actual parameter θ.

There are two main branches in estimator theory: We need different methods for finding estimators, and we need to evaluate how accurate the estimators are.

There are also two types of estimators: Point estimators and interval estimators.

3.1 Finding Point Estimators

Definition 3.1.1 (Point Estimators)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $I \subseteq \mathbb{R}$ be an indexing set. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. A point estimator of the random sample is a function $\widehat{\theta} \circ X_\theta : \Omega \to I$.

Note: This the same definition of a statistic.

Definition 3.1.2 (Method of Moments)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $J_1, \dots, J_k \subseteq \mathbb{R}$ be indexing sets. Let $\{D_{\theta_1,\dots,\theta_k} \mid \theta_i \in J_i\}$ be a family of distributions. Let $X_{\theta_1,\dots,\theta_k} : \Omega \to \mathbb{R}^n$ be a random sample with distribution $D_{\theta_1,\dots,\theta_k}$ and probability distribution function $f_{\theta_1,\dots,\theta_k}$. Define the methods of moments estimator to be the statistics $\widehat{\theta}_1, \dots, \widehat{\theta}_k$ given by the solving the system of equations

$$\frac{1}{n}\sum_{j=1}^{n}(X_{\theta_1,\dots,\theta_k})_j^i = E[D^i_{\widehat{\theta}_1,\dots,\widehat{\theta}_k}]$$

for $1 \leq i \leq k$.

Example 3.1.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample with normal distribution $X_i \sim \text{Normal}(\theta_1, \theta_2)$. The method of moments estimator for X is given by solving

$$\frac{1}{n}\sum_{j=1}^{n}(X_{\theta_1,\dots,\theta_k})_j = E[\text{Normal}(\widehat{\theta}_1, \widehat{\theta}_2)] \quad \text{and} \quad \frac{1}{n}\sum_{j=1}^{n}(X_{\theta_1,\dots,\theta_k})_j^2 = E[\text{Normal}(\widehat{\theta}_1, \widehat{\theta}_2)^2]$$

which we may rewrite as

$$\overline{X_{\theta_1,\dots,\theta_k}} = \widehat{\theta}_1 \quad \text{and} \quad \frac{1}{n}\sum_{j=1}^{n}(X_{\theta_1,\dots,\theta_k})_j^2 = \widehat{\theta}_1^2 + \widehat{\theta}_2^2$$

Omitting the dependency of the random variables on the parameters, we get

$$\widehat{\theta}_1 = \overline{X} \quad \text{and} \quad \widehat{\theta}_2 = \frac{1}{n}\sum_{j=1}^{n}(X_j - \overline{X})^2$$

We are lucky in the above example because the estimator accurately recovers the actual parameters. But in general, this is not true.

Example 3.1.4
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample with binomial distribution $X_i \sim \text{Binom}(\theta_1, \theta_2)$. The method of moments estimator for X is given by solving

$$\frac{1}{n}\sum_{j=1}^n X_j = E[\text{Binom}(\widehat{\theta}_1, \widehat{\theta}_2)] \quad \text{and} \quad \frac{1}{n}\sum_{j=1}^n X_j^2 = E[\text{Binom}(\widehat{\theta}_1, \widehat{\theta}_2)^2]$$

which we may rewrite as

$$\overline{X} = \widehat{\theta}_1\widehat{\theta}_2 \quad \text{and} \quad \frac{1}{n}\sum_{j=1}^n X_j^2 = \widehat{\theta}_1\widehat{\theta}_2(1-\widehat{\theta}_2) + \widehat{\theta}_1^2\widehat{\theta}_2^2$$

Solving the simultaneous equation gives

$$\widehat{\theta}_1 = \frac{\overline{X}^2}{\overline{X} - \frac{1}{n}\sum_{j=1}^n (X_j - \overline{X})^2} \quad \text{and} \quad \widehat{\theta}_2 = \frac{\overline{X} - \frac{1}{n}\sum_{j=1}^n (X_j - \overline{X})^2}{\overline{X}} = \frac{\overline{X}}{\widehat{\theta}_1}$$

Definition 3.1.5 (Maximum Likelihood Estimators)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $I \subseteq \mathbb{R}$ be an indexing set. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. Let $\widehat{\theta}$ be a point estimator for the random sample. We say that $\widehat{\theta}$ is a maximum likelihood estimator if there exists $x \in \mathbb{R}^n$ such that

$$L(\widehat{\theta}(x))(x) = \max\{L(\theta)(x) : \theta \in I\}$$

3.2 Evaluating Point Estimators

Definition 3.2.1 (Bias of a Point Estimator)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. Define the bias of $\widehat{\theta}$ to be

$$\text{Bias}(\widehat{\theta}) = E[\widehat{\theta}] - \theta$$

Definition 3.2.2 (Unbiased Point Estimators)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. We say that $\widehat{\theta}$ is unbiased if $\text{Bias}(\widehat{\theta}) = 0$.

Definition 3.2.3 (Mean Squared Error)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. Define the mean squared error of $\widehat{\theta}$ to be

$$\text{MSE}(\widehat{\theta}) = E[(\widehat{\theta} - \theta)^2]$$

Theorem 3.2.4 (The Bias Variance Trade Off)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X_1, \dots, X_n : \Omega \to \mathbb{R}$ be a random sample. Let $\widehat{\theta}$ be a point estimator of the random sample. Then we have

$$\text{MSE}(\widehat{\theta}) = \text{Var}(\widehat{\theta}) + (\text{Bias}(\widehat{\theta}))^2$$

3.3 Finding Interval Estimators

3.4 Evaluating Interval Estimators

4 Hypothesis Testing

4.1 Finding Hypothesis Tests

Definition 4.1.1 (Hypothesis)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $\{D_\theta \mid \theta \in I\}$ be a family of distributions. Let $X_\theta : \Omega \to \mathbb{R}^n$ be a random sample with distribution D_θ. A hypothesis is s subset of I.

4.2 Evaluating Hypothesis Tests

5 Regression Analysis

Definition 5.0.1 (Linear Regression Space)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample such that $E[X] < \infty$. Define the linear regression space of the sample to be

$$\mathcal{H}(X) = \left\{ \sum_{i=1}^{n} \beta_i X_i + \alpha \;\middle|\; \alpha, \beta_i \in \mathbb{R} \right\} \subseteq L^2(\Omega)$$

Definition 5.0.2 (Linear Regression)
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample. Let $Y : \Omega \to \mathbb{R}^n$ be a random sample. Let $L : \mathcal{H} \to \mathbb{R}$ be a loss function. The linear regression of Y on X with L is a random variable $\widehat{Y} \in \mathcal{H}(X)$ such that

$$L(\widehat{Y}) = \min\{L(Z) \mid Z \in \mathcal{H}(X)\}$$

Proposition 5.0.3
Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $X : \Omega \to \mathbb{R}^n$ be a random sample. Let $Y : \Omega \to \mathbb{R}^n$ be a random sample. Then the linear regression of Y on X with the mean squared error exists and is unique.